

Drake AI Enterprise Platform

Clean technical stack summary for modules, result models, and UI.

Project Overview

Drake AI is a full-stack petroleum intelligence platform. The frontend is a React/Vite web app. The backend is a FastAPI Python API. Data workflows include LAS processing, petrophysics, seismic enhancement, CCUS screening, geothermal screening, production intelligence, reports, and an AI copilot.

Layer	Technology Used
Frontend	React 18, TypeScript, Vite, Tailwind CSS, React Router, Zustand, TanStack Query
Backend	FastAPI, Python, Pydantic, SQLAlchemy, Alembic
Database	SQLite for local default, PostgreSQL 15 in Docker
Cache	Redis for session/cache support, with local fallback
Storage	Local uploads plus MinIO/S3-compatible configuration
Containers	Docker and Docker Compose
Visualization	Plotly, Recharts, custom React tables and dashboards
AI/ML	scikit-learn, XGBoost optional, TensorFlow/Keras, PyWavelets, NumPy, pandas, SciPy

Models Used For Results

Result Area	Model / Method Used
Drake GPT Copilot	Anthropic Claude Sonnet 4: claude-sonnet-4-20250514. Local rule-based fallback when no API key is configured.
Missing Log Prediction	Extra Trees, Random Forest, and Gradient Boosting regressors.
AI Parameter Prediction	Empirical formulas, Random Forest AI, XGBoost AI, Gradient Boosting AI, and Decision Tree AI.
AI Uncertainty	P10/P50/P90 bands from Random Forest tree spread, bootstrap/ensemble spread, or percent-based fallback.
Seismic Enhancement	Conv1D U-Net, PyWavelets + Random Forest, Conv1D frequency-band autoencoder, and FFT fallback.
CCUS Screening	Rule-based reservoir/seal quality scoring from log-derived PHIE, VSH, permeability, GR, RHOB, NPHI, and RT.
Geothermal / Production	Analytical calculations and KPI rules from uploaded data.

Module Stack Usage

Module	Stack / Usage
Auth	FastAPI, JWT, bcrypt, SQLAlchemy user records.
Projects / Wells / Curves / Files	FastAPI APIs, SQLAlchemy, LAS parsing, local uploads, optional MinIO.
Petrophysics	lasio, pandas, NumPy, SciPy, Plotly JSON, Redis-backed sessions.
Missing Logs	scikit-learn regressors with LAS feature inputs and CSV/LAS exports.
AI Prediction / Uncertainty	Random Forest, XGBoost, Gradient Boosting, Decision Tree, empirical petrophysics formulas.
Facies / Formation Tops	Petrophysics toolbox logic, clustering/classification style workflows, Plotly responses.
Seismic	segio, TensorFlow/Keras, PyWavelets, Random Forest, FFT, NumPy.
CCUS	Custom LAS parser, cutoff rules, quality scoring, Plotly log tracks, XLSX export.
Geothermal	LAS screening, pandas, CSV/JSON export, heat-flow map PNG.
Production	CSV/XLSX ingestion, pandas analytics, production KPI outputs.
Reports	PDF, spreadsheet, CSV/JSON export utilities.
Standalone Drake_Uncertainty	Flask, templates, Plotly, lasio, pandas, scikit-learn, optional XGBoost.

Complete UI Technical Stack

UI Part	Technology
App shell	React 18, TypeScript, Vite, React Router DOM
State and data	Zustand, TanStack Query, Axios, local/session storage
Styling	Tailwind CSS, PostCSS, Autoprefixer, custom CSS and inline component styles
Charts	Plotly.js and Recharts
Uploads	React Dropzone and typed API service wrappers
Notifications and icons	React Hot Toast and lucide-react
Main screens	Dashboard, projects, data management, petrophysics, seismic, production, CCUS, geothermal, reports, settings

Main API Groups

/api/auth, /api/projects, /api/wells, /api/curves, /api/files, /api/ai, /api/reports, /api/gpt, /api/petrophysics, /api/seismic, /api/geothermal, /api/production, and /api/ccus.